

Figure 5: Algorithm of Annealing

Input: P_{original}
Output: S_{optimal}

Function SolveQUBOviaQuantumAnnealing(P_{original}):

```
// 1. Problem Transformation
C ← InitializeEmptyQUBOMatrix
Q ← InitializeemptyQUBOMatrix
for each variable v in P_original do
    MapToABinaryVariable(v, C, {0,1})
end

// QUBO Q
ConvertObjectiveFunction(P_original) to  $\sum_{i \leq j} Q_{ij} x_i x_j$ 

// QUBO Q
for each constraint c in P_original do
    PenaltyTerm_c ← FormulateQuadraticPenaltyForc
    AddPenaltyTerm_c to  $\sum_{i \leq j} Q_{ij} x_i x_j$ 
end

H_problem ← ConstructQUBOHamiltonianFromQ // H_problem

// 2. Physical Solution
H_initial ←
    DefineInitialSimpleHamiltonian(e.g., transverse field) //
H_total(t) = (1 - t/T_a) H_initial + (t/T_a) H_problem //
InitializeQuantumSystemInGroundStateOf H_initial //

for t ← 0 to T_a, Δt do
    ApplyH_total(t) toQuantumSystem
    AllowSystemToEvolveAdiabatically
end

// 3.
Measurement ← MeasureQuantumSystemInComputationalBasis
S_binary ←
    ExtractBinaryValues{x_1, x_2, ..., x_N} fromMeasurement //
S_optimal ← MapS_binary backToP_original variables
return S_optimal
```